

Isaac Biggs

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EDUCATION

University of Illinois Urbana-Champaign

Expected: May 2028

Bachelor of Science in Computer Science + Geography & GIS | Data Science Minor

GPA: 3.86/4.00

- **Relevant Coursework:** Data Structures & Algorithms, Principles of GIS, Computer Systems, Database Systems

SKILLS

- **Languages & Frameworks:** Python, SQL, C++, Java, C, R, JavaScript, HTML/CSS, React, Flask
- **GIS & Mapping:** ArcGIS (Pro, Enterprise, Field Maps), ArcPy, GeoPandas, Shapely, Rasterio, geopy, Leaflet
- **Data Analysis:** Tableau, Excel, Jupyter Notebook, Matplotlib
- **Databases & Tools:** PostgreSQL/PostGIS, Docker, Git/GitHub, SQLite, VS Code
- **Machine Learning:** PyTorch, scikit-learn, XGBoost
- **Additional Tools:** AutoCAD, Microsoft Office, Photoshop, Premiere Pro, Blender

WORK EXPERIENCE

Geospatial Software Engineer Intern, Applied Pavement Technology – Champaign, IL

May 2025 - Present

- Building a ML model using PyTorch to automatically align and register aerial imagery, improving georeferencing accuracy and streamlining image processing workflows to enhance company operational efficiency
- Developing ArcGIS Pro scripts using Python, ArcPy, and Shapely to automate spatial joins, buffering, and network analysis, replacing manual workflows and reducing processing times from days to minutes
- Authoring in-depth technical documentation for technicians unfamiliar with scripting, and iterating on tool design based on engineer and technician feedback to align functionality with workflows and goals
- Designing scripts as reusable tools, deployable across large-scale geospatial datasets spanning airports, roadways, and military installations to support long-term consulting operations beyond a single project scope

Data Analyst Intern, Applied Technologies for Learning in LAS – Champaign, IL

Aug 2025 - Dec 2025

- Analyzed IT Help Desk ticket data using Excel, Tableau, and Python (pandas), geocoding tickets by department to visualize workload distribution spatially and identify trends in response times across service areas
- Delivered Tableau dashboards to managers, providing data-driven recommendations for resource allocation and staffing based on trends, enabling managers to reallocate 10+ staffing hours per week to high-traffic periods

Course Assistant, Introduction to Computer Science (CS124) – Champaign, IL

Jan 2025 – Dec 2025

- Tutored students in foundational Java programming through one-on-one help site, assisting with debugging, assignments, and quizzes while collaborating with instructors and mentoring new assistant tutors

RELEVANT PROJECTS

Satellite Imagery Natural Disaster Assessment | Python, PyTorch, Rasterio, Shapely

July 2026

- Developed PyTorch-based preprocessing pipeline for disaster damage dataset (45K+ images), implementing vector-to-raster conversion with Shapely/Rasterio and intelligent tiling with overlap
- Architected dual-task PyTorch Dataset supporting semantic segmentation and change detection with joint augmentation, comprehensive testing (23 unit tests), and configuration-driven workflows

Transit Delay Predictor | Python, SQLite, Pandas, XGBoost, NumPy, geopy, scikit-learn

June 2026

- Built a geospatial ML pipeline (XGBoost, scikit-learn) on Chicago Transit Authority GTFS data with SQLite persistence, engineering weather and spatial features to generate 13,800+ delay predictions across 144 stops
- Deployed an interactive MapLibre GL JS web map visualizing those predictions by month and time of day

Chicago Crime Analytics Web App | Python, React, Flask, GeoPandas, Pandas, Geopy

May 2025

- Built an interactive Flask + React dashboard analyzing 100K+ Chicago crime reports alongside demographic datasets, letting users query by location, time, and crime type to surface neighborhood-level risk exposure
- Cut data processing time 70% via optimized Pandas indexing and vectorized querying, enabling real-time visualization across 15+ datasets that gave non-technical users data-driven insight into community safety trends

LEADERSHIP AND INVOLVEMENT

Chapter President, Phi Sigma Kappa – Champaign, IL

December 2025 - Present

- Lead chapter of 100+ other undergraduate members to promote brotherhood, scholarship, and character
- Serve as the main liaison between the chapter, alumni board, national headquarters, other Greek organizations, and UIUC Fraternity and Sorority Affairs to facilitate the health and success of our chapter
- Manage a \$450,000 yearly budget while guiding financial strategy, transparency, and long-term sustainability